

WEEK 09 TASK: PROGRESS AND INTEGRATION REPORT FEATURE COMPLETION, INTEGRATION & MID-PROJECT REVIEW

TEAM NAME: CSE4204-8D-T01 | **SECTION:** 8D

Project Title: NUBTK_COPILOT - AI-Powered University ERP and Career Architect

Instructor: Md. Riaz Mahmud, Assistant Professor, Dept of CSE, NUBTK

Team ID: CSE4204-8D-T01

GitHub Repository: <https://github.com/bashirnubtk/cse4204-8d-t01-nubtk-copilot>

1. Team Information & Core Engineering Roles

Full Member Name	Student ID	System Design & Engineering Responsibilities
Bashir Alam	11220320963	Team Leader. Core backend architecture, OpenRouter Hybrid AI Copilot Engine (bot_engine.py), Scikit-Learn ML models (models_ml.py), Keras Neural Network (neural_net.py), REST API ViewSets, and Full-system Integration.
Tisha Rani Chakroborty	11220320954	Frontend Developer. Dynamic Student Analytics UI/UX components, Real-time AI Workspace views, Jinja2/Tailwind CSS integration, and Mobile responsiveness.
Fariha Rahman Oishee	11220320967	Database Manager. PostgreSQL schema extension for Courses, Student Results, Resource Hubs, AI Chat Sessions, and database queries optimization.
Ratul Raha Shuvo	11220320949	Backend Support. ReportLab PDF automated generation utilities, NUBTK 160-credit CSE curriculum seeder, and error handling middleware.

2. Project Objective & Scope Overview

The primary objective of Week 09 development cycle is to transition NUBTK_COPILOT from individual functional components into a fully integrated, cohesive, end-to-end operational software ecosystem. All initial backend endpoints, database models, frontend interfaces, Scikit-Learn predictive algorithms, Keras neural networks, and OpenRouter multi-model LLM features have been connected and tested on real application workflows.

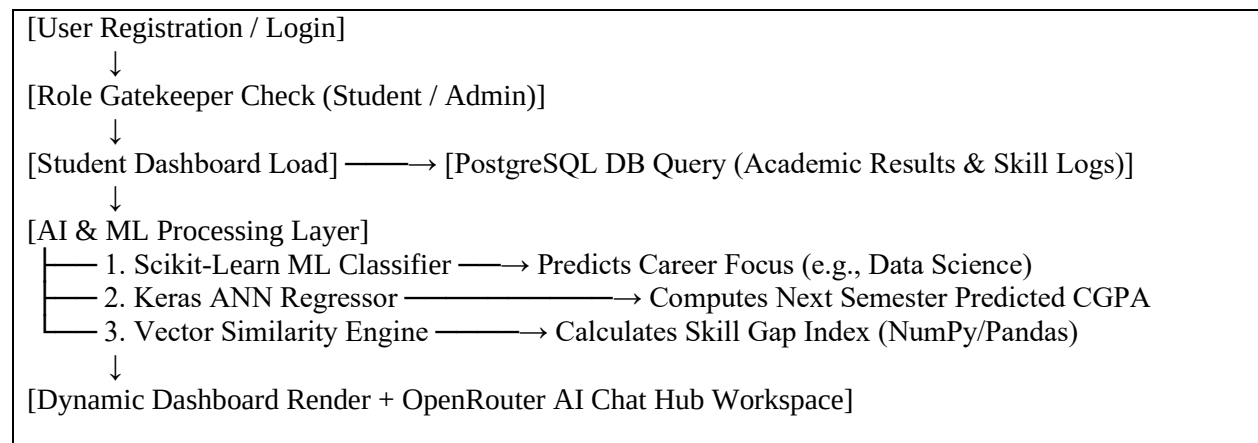
3. Feature Completion Checklist

Feature Name / Module	Status	Reason (If Incomplete) & Planned Completion
User Registration & Role-Based Gateways	Completed	Fully integrated with Django session-based authentication.
Student Dashboard & Academic Performance View	Completed	Renders subject-wise marks, semester CGPA, and automated digital ID.
Scikit-Learn ML Career Track Classifier (Decision Tree / SVM)	Completed	Live vector classification across Software Engineering, Data Science & Cyber Security.
Keras Neural Network (3-Layer ANN CGPA Regressor)	Completed	Forecasts CGPA based on attendance, study hours, and quiz averages.
OpenRouter Multi-Model AI Chat Hub (bot_engine.py)	Completed	System prompt context injection active with automated multi-LLM fallback logic.
Admin AI Control Center & University Seeder	Completed	Full institutional analytics, student approvals, and 160-credit CSE curriculum seeding.
ReportLab Digital ID & Receipt PDF Generator	Completed	Automated NUBTK token generation and PDF export.
Real-Time Push Notifications (WebSocket/Celery)	In Progress	Currently relying on HTTP long-polling/AJAX. Full WebSocket implementation scheduled for Week 11.

4. Integration Status Summary

- **Frontend–Backend Integration:** Standardized across Django Views, Jinja2 Templates, and REST API ViewSets using JSON payload exchanges. Hardcoded/dummy data replaced with live database objects.
- **AI Integration:** Integrated into real workflows. Contextual data (Student CGPA, Skill Vectors, Course History) is fetched from PostgreSQL and injected into system prompts before routing through OpenRouter API gateways.
- **Authentication & Security:** Complete role-based authentication separating Students and Administrators. API credentials and secrets are managed via environment variables.

5. End-to-End Major User Workflows



6. Error Handling & Robustness Implementations

- **OpenRouter API Failures / Rate Limits:** Implemented explicit try-except fallback handlers in `bot_engine.py` that cycle sequentially through models (Gemini Flash, Llama 3.1, DeepSeek R1, OpenRouter Auto) if HTTP 400/404 errors occur.

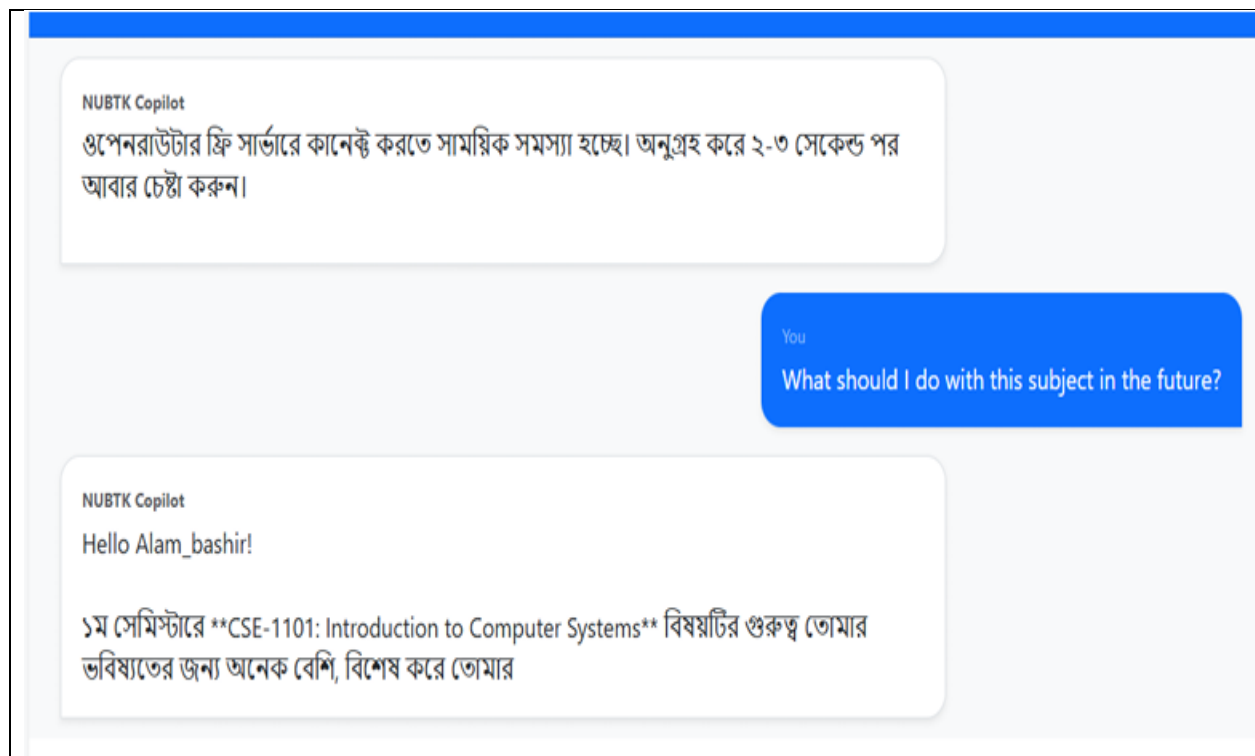
- **Invalid Form Inputs:** Form validation implemented on frontend/backend to catch missing fields and display user-friendly error messages without breaking the execution flow.
- **Database Inaccessibility:** Graceful error boundaries implemented in views to display empty state alerts rather than server 500 exceptions.

7. Current System Screenshots & Placement Instructions

Below are the specific locations where system screenshots must be attached in your final submission document:

7.1 Student AI Workspace & Context-Aware Chat

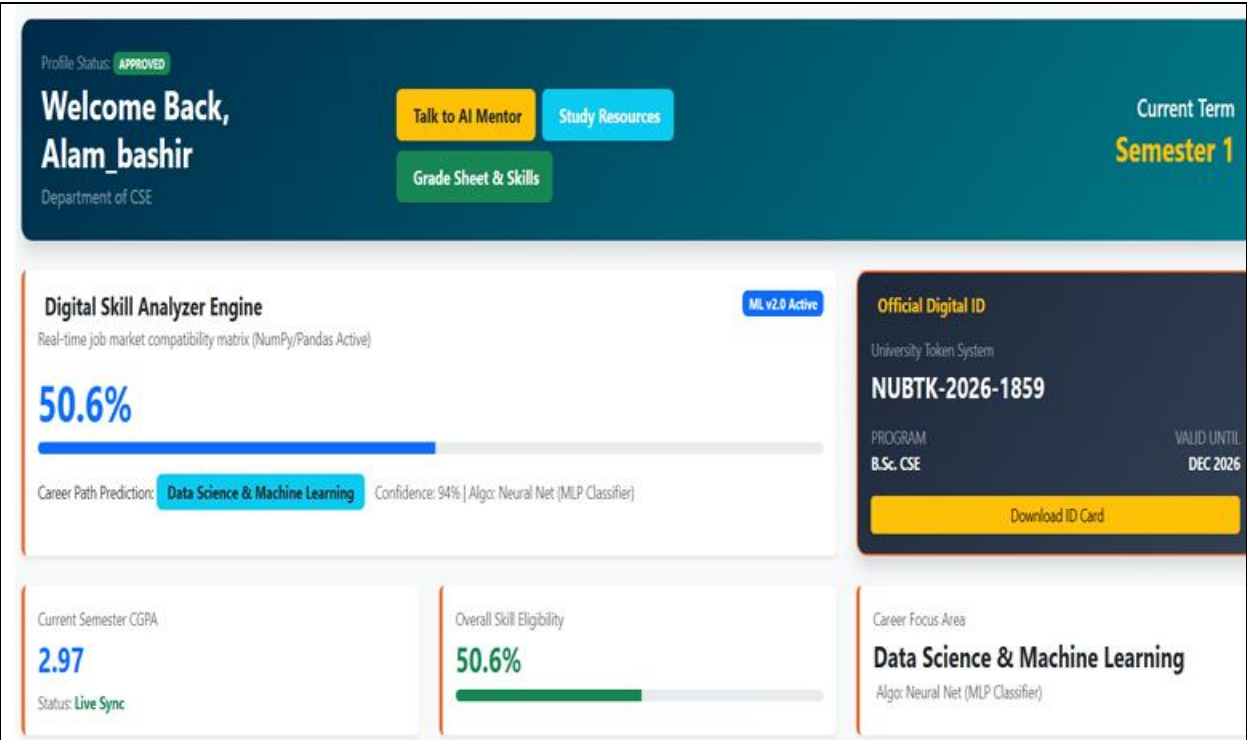
Description: Shows real-time AI Chat Hub conversation where student queries receive personalized, context-aware responses, as well as backend console execution showing OpenRouter model handling and automated fallbacks.



[AI Chat Hub UI & OpenRouter Terminal Execution Log]

7.2 Dynamic Career Path Prediction & Student Dashboard

Description: Displays the main Student Dashboard rendering live student CGPA, Digital Skill Analyzer Engine score (50.6%), Official Digital ID (NUBTK-2026-1859), and Scikit-Learn/ANN Career Focus prediction.



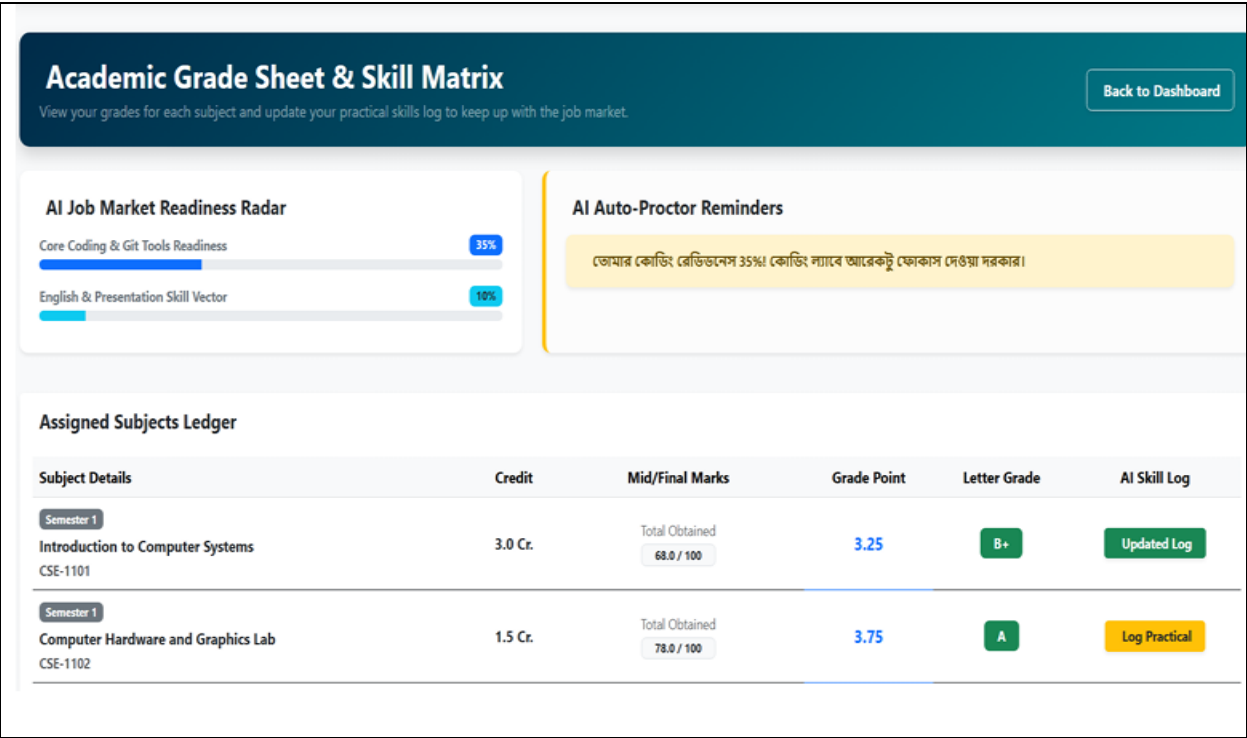
[Student Dashboard & Skill Analyzer Engine]

7.3 Academic Grade Sheet & Skill Matrix

Description: Shows subject-wise breakdown (Class Tests, Quizzes, Attendance, Subject CGPA) along with AI Job Market Readiness Radar and Auto-Proctor Reminders.

Subject-Wise Marks & Performance Matrix						Update Skill Logs
Course Name / Code	Class Test (30)	Quiz (20)	Attendance %	Final Exam	Subject CGPA	
Introduction to Computer Systems (CSE-1101)	10.00	5.00	6.00%	47.00	3.25	
Computer Hardware and Graphics Lab (CSE-1102)	10.00	0.0	8.00%	60.00	3.75	
Structured Programming Language (CSE-1103)	5.00	5.00	3.00%	57.00	3.50	
Structured Programming Lab (CSE-1104)	0.0	0.0	0.0%	50.00	2.50	
Differential and Integral Calculus (MATH-1101)	5.00	5.00	4.98%	40.00	2.50	
English Composition and Communication (ENG-1101)	5.00	5.00	5.00%	30.00	2.25	
Physics (Waves, Optics and Thermodynamics) (PHY-1101)	4.00	5.00	10.00%	50.00	3.25	
Physics Lab (PHY-1102)	7.00	4.00	7.00%	40.00	2.75	

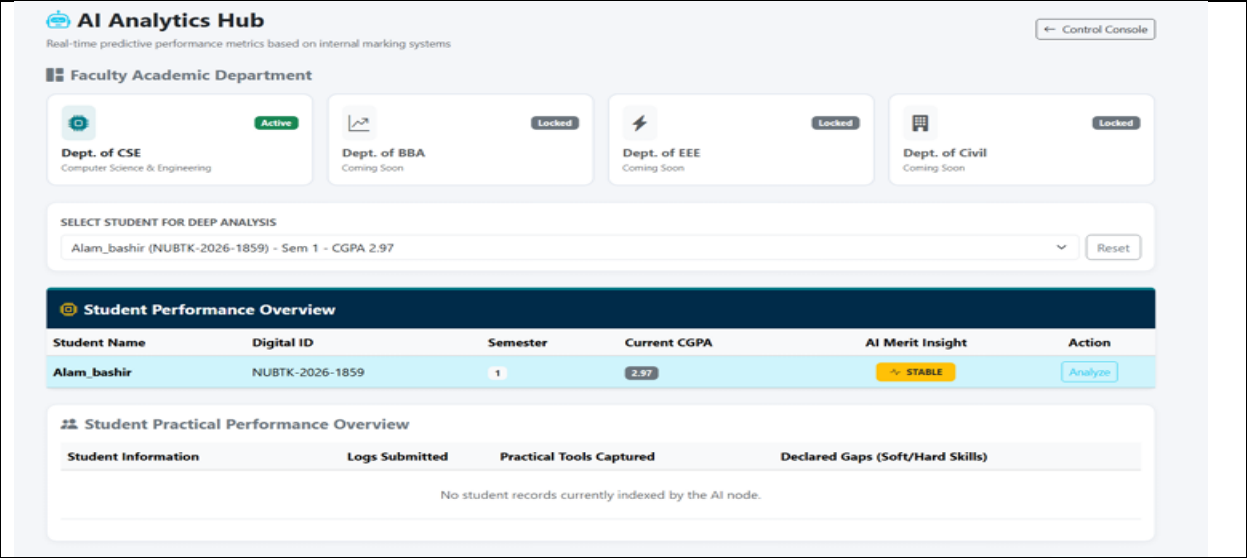
Northern University of Business and Technology, Khulna
© 2026 Department of Computer Science & Engineering. Developed under CSE 4204 Project Frameworks.



[Grade Sheet & Skill Matrix Workspace]

7.4 Admin AI Analytics Hub & Departmental Control

Description: Displays the institutional administration interface with deep analysis of student merit insights, neural network future projections chart, critical skill gaps, and approval controls.





[Admin AI Analytics Dashboard & Keras Projections Chart]

7.5 GitHub Repository & Multi-Branch Activity

Description: Captures the active GitHub repository branch structure (main, bashir_core_backend, tisha_frontend, dev, oishee_database, ratul_ui_frontend) demonstrating continuous team contributions.

Your branches					
Branch	Updated	Check status	Behind	Ahead	Pull request
tisha_frontend	4 days ago		19	9	
bashir_core_backend	last week		2	8	
dev	2 months ago		2	3	
oishee_database	2 months ago		19	2	
ratul_ui_frontend	2 months ago		19	2	

Active branches					
Branch	Updated	Check status	Behind	Ahead	Pull request
tisha_frontend	4 days ago		19	9	
bashir_core_backend	last week		2	8	
dev	2 months ago		2	3	
oishee_database	2 months ago		19	2	
ratul_ui_frontend	2 months ago		19	2	

[GitHub Repository Branches View]

8. GitHub Repository & Commit Policy Compliance

Repository Link: <https://github.com/bashirnubtk/cse4204-8d-t01-nubtk-copilot>

Recent Standardized Commits:

- feat(integration): completed frontend-backend integration for student dashboard
- feat(ai_engine): finalized openrouter system prompt context injection & fallback logic
- feat(ml_analytics): connected scikit-learn career classifier with live database vectors
- feat(neural_net): validated keras ann cgpa regressor on real academic logs
- fix(auth): secured protected routes and optimized session handling
- docs(report): updated week 09 progress report and integration architecture

9. Deliverables & Links

- **GitHub Repository:** <https://github.com/bashirnubtk/cse4204-8d-t01-nubtk-copilot>
- **Live Demo URL:** <http://nubtk-copilot.local> (Local Development / Staging Server)
- **Code Dump Utility:** code_dumper.py configured for standardized project export.